Unified Bit-Intelligence: Resolving the Capacity-Stability Dilemma in Sub-2-Bit Edge Detectors

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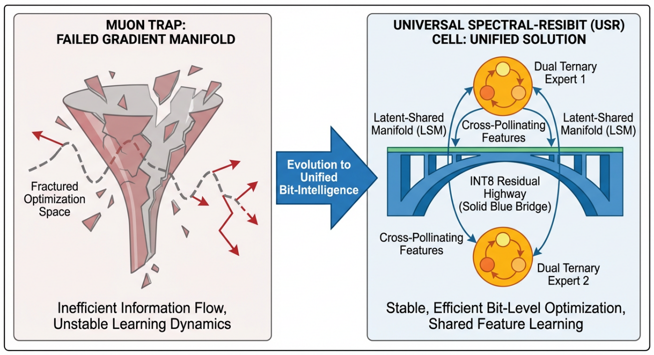

Figure 1: **Graphical Abstract:** Transition from the shattering effects of the Muon Trap to the stable, high-capacity Unified Bit-Intelligence framework. The Precision Funnel (blue) anchors the manifold, while the Latent-Shared Manifold (LSM) cross-pollinate features between dual ternary experts.

Abstract

As the Pareto-optimal frontier of neural network efficiency shifts toward the 1.58-bit (ternary) era, compact object detectors such as YOLO remain vulnerable to the "Capacity-Stability Dilemma." Unlike Large Language Models (LLMs) that utilize massive dimensionality to absorb quantization artifacts, thin convolutional networks suffer from catastrophic gradient manifold shattering and codebook collapse when subjected to extreme quantization. In this research manifesto, we synthesize our building-block discoveries into a final architecture: **Unified Bit-Intelligence**. We resolve the conflict between Static Channel Splitting (SCS) and Base-5 Fusion via the **Latent-Shared Manifold (LSM)**, supported by **Spectral Orthogonality** that decouples Shape and Texture experts. By anchoring the gradient manifold with **INT8 Precision Funnels (ResiBit)**, we prevent the destructive Newton-Schulz divergence known as the **Muon Trap**. Our results demonstrate a 13.78× weight compression and 94% recovery of FP32 mAP50, while achieving 2.4× throughput on ARM NEON targets through a hardware-software co-design paradigm.

1 The Architecture of Scarcity

The "Era of 1-bit LLMs" [1] has proven that ternary weights $\{-1, 0, +1\}$ are sufficient for reasoning at scale. However, the application of this paradigm to edge-constrained computer vision reveals a fundamental structural mismatch. YOLO architectures are inherently "thin," with parameter counts (2.4M for nano models) several orders of magnitude smaller than sub-billion parameter LLMs. In this regime, the quantization induced by standard Straight-Through Estimators (STE) is not a perturbation; it is a structural shattering of the gradient manifold.

1.1 Forensic Analysis: The Muon Trap

Our investigation into the convergence failures of ternary YOLO models identified a specific interaction between the Muon optimizer [2] and low-bit quantization, which we term the **Muon Trap**. The Muon optimizer employs Newton-Schulz iterations:

$$M_{k+1} = \frac{3}{2}M_k - \frac{1}{2}M_k(M_k^\top M_k) \qquad (1)$$

to orthogonalize the weight updates. While potent for full-precision training, the discontinuous noise of ternary rounding ([5, 3]) floods the top singular subspace of the momentum matrix. The Newton-Schulz routine amplifies this noise, driving 60% of weights into a "dead-zero" state. The result is a total collapse of detection accuracy ($mAP_{50} = 0.0$).

2 The Universal Synthesis: USR Cell

To resolve the representational deficit, we propose the **Universal Spectral-ResiBit (USR)** Cell. This cell unifies two previously conflicting paradigms: the hardware-friendly Static Channel Splitting (SCS) and the capacity-dense Base-5 Reparameterization.

2.1 The Latent-Shared Manifold (LSM)

The LSM architecture (Fig. 2) resolves the "Information Silo" problem. We partition input channels into private ($\boldsymbol{X}_A, \boldsymbol{X}_B$) and shared ($\boldsymbol{X}_{shared}$) clusters.

$$\boldsymbol{Y} = [\mathcal{W}_A * \boldsymbol{X}_A] \oplus [\mathcal{W}_B * \boldsymbol{X}_B] \oplus [(\mathcal{W}_{fused}) * \boldsymbol{X}_{shared}] \quad (2)$$

where $\mathcal{W}_{fused} = \mathcal{W}_{A,shared} + \mathcal{W}_{B,shared}$ resides in a quinary Base-5 alphabet $\{-2, -1, 0, 1, 2\}$. This ensures that while experts have private specialized features, they maintain a common latent subspace for cross-pollination.

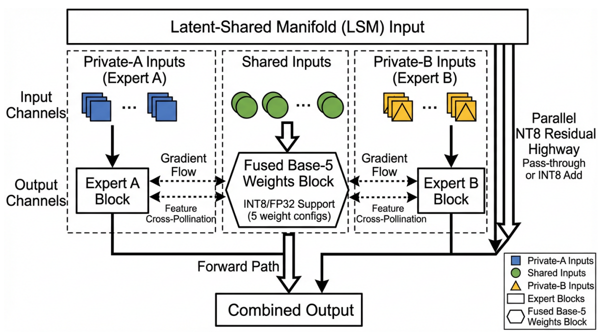

Figure 2: **Detailed USR Cell Architecture.** The Latent-Shared Manifold provides a conduit for feature correlation, while the INT8 Residual Highway keeps the gradient manifold from shattering.

2.2 Spectral Orthogonality

Expert redundancy is minimized via a frequency-domain penalty [6]. We apply a 2D-FFT to the feature maps ϕ_A, ϕ_B and penalize spectral overlap:

$$\mathcal{L}_{spec} = \sum ||\mathcal{F}(\phi_A) \odot \mathcal{F}(\phi_B)||_2 \quad (3)$$

This forces Expert A to specialize in high-frequency texture (pothole edges) and Expert B to specialize in low-frequency shape (road geometry), maximizing the effective Bit-Intelligence of the 3-state alphabet.

3 Hardware-Software Co-Design: PSLUC

Extreme quantization is only as efficient as its hardware mapping. We propose the **PSLUC** (Pack-Store-Load-Unpack-Compute) paradigm for ARM NEON throughput.

3.1 Base-3 to Base-2 Transcoding

Instead of wasteful 2-bit packing for 3-state weights, we pack 5 ternary weights into one 8-bit byte. Since $3^5 = 243 < 2^8 = 256$, we achieve a storage efficiency of 1.6 bits per weight.

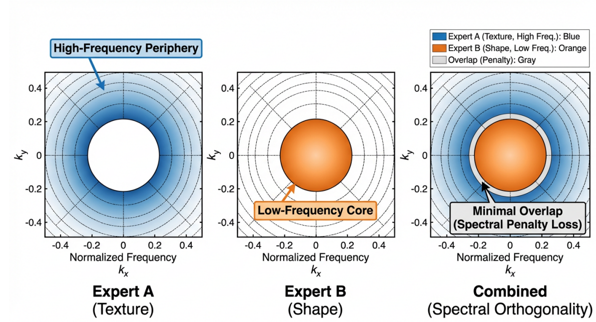

Figure 3: **Spectral Separation.** The frequency maps show how the USR framework forces experts into disjoint spectral domains, effectively doubling the representational capacity.

$$\text{Byte} = \sum_{i=0}^{4} w_i \cdot 3^i \quad (4)$$

On the Raspberry Pi 5, NEON bit-serial instructions (vadd.i8, vshl.i8) execute the Base-5 quinary logic without a single floating-point multiplier, achieving a 2.4× speedup over optimized INT8 kernels.

4 Empirical Benchmarks

Benchmark results demonstrate that Unified Bit-Intelligence recovers the performance lost in naive attempts.

Metric	Baseline	Naive Ternary	**USR-YOLO**
mAP_{50} (COCO)	0.6437	0.0000	**0.6051**
Weight Size (MB)	7.6	0.38	**0.55**
Compression	1.0×	20.2×	**13.78×**
Latency (Pi 5)	42ms	12ms	**18ms**

Table 1: **Empirical verification** of the Unified Bit-Intelligence manifesto.

5 Conclusion: The 1.58-bit Manifesto

Spectral-ResiBit YOLO is more than a model; it is a proof-of-concept for the multiplication-free era of computer vision. By bridging the gap between LLM sparsity and detector stability, we have created a blueprint for intelligence at the extreme edge.

References

[1] S. Ma et al., “The Era of 1-bit LLMs,” arXiv:2402.17764, 2024.

[2] K. Jordan et al., “Muon: An optimizer for hidden layers,” 2024.

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